A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the Mathematics B Paper 1 question papers and mark schemes. Methods are laid out the way the mark schemes award M, A and B marks, and every warning comes from a mark-scheme note such as “working required”, “do not ISW”, “SC” or “correct answer from obvious incorrect working scores 0”. The formulae sheet is printed in the paper; everything else must be learnt.

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Papers analysed: Jan 2019 (legacy 4MB0), Jun 2019, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 How Paper 1 is built

Format: 1 hour 30 minutes, 100 marks, 25–29 short questions (each 1–6 marks), calculator allowed. That is under 1 minute per mark, so speed and accuracy matter. Unlike Mathematics A, there is **no formulae sheet**, and Mathematics B includes **matrices**, **ruler-and-compass loci**, **symmetry** and **formal geometric proofs** (congruence and similarity with reasons). “You must write down all the stages in your working.”

Topic	Papers (of 16)	What is asked most
Surds (without a calculator)	15	simplify $\sqrt{675} - \sqrt{108}$; rationalise $(a + \sqrt{b})/(c - \sqrt{d})$
Fractions, HCF/LCM, prime factors	14	mixed-number arithmetic shown; LCM of three numbers; remainders
Algebraic fractions and factorising	15	simplify a quotient of quadratics; single fraction; subject
Percentages and ratio	15	reverse %, repeated change, three-part ratios
Bounds	13	upper/lower bound of speed, time, density, angle
Circle theorems and angle reasons	13	find an angle with reasons; intersecting chords/secants
Probability (trees, conditional)	13	complete a tree; exactly one; unknown counts
Statistics	13	grouped mean, histogram, combined means, find a missing value
Constructions and loci	11	perpendicular bisector, angle bisector, arcs, region R
Proportion	11	direct/inverse, square, cube, root; chained proportion
Matrices	11	A + kB, AB, determinant, inverse, unknown elements
Similar shapes (area/volume)	10	lengths from area or volume ratios; percentage changes
Quadratics: complete the square, inequalities	10	$a + b(x + c)^2$, range of a function, set of values
Congruence/similarity proofs	8	SAS, ASA, RHS, SSS with reasons; similar by AA

Topic	Papers (of 16)	What is asked most
Calculus and kinematics	8	dy/dx, stationary points, speed–time graphs
Sets and Venn diagrams	9	list elements, n(A ∩ B′), 3-set Venn from statements

2 TWENTY MARK-SCHEME RULES

1. **"Without a calculator": correct answer with no working scores no marks.** Show every decomposition: √675 = √(225 × 3) must be seen (Jun 25).
2. **Numerical checks are not proofs:** "numerical approaches (use of p = 1.618) are not acceptable" (Jun 22); "no marks if numerical examples considered only" (Jan 21).
3. **Fractions "show that":** write the improper fractions and the common-denominator step; a calculator answer alone scores nothing.
4. **Proofs need the key words:** congruence and similarity marks need each pair of equal sides/angles **with a reason** ("opposite sides of a parallelogram are equal", "angles in the same segment") and the condition (SAS, ASA, AAS, RHS, SSS).
5. **Angle reasons:** at least the underlined words ("angle at centre is twice angle at circumference", "alternate segment theorem"); one of them must be a circle theorem where relevant. Assuming a triangle is isosceles loses the mark.
6. **Constructions:** arcs must be visible (two pairs of intersecting arcs of equal radius); a line without arcs scores at most B1. Lines must lie within the overlay tolerance.
7. **Region R:** only awarded if all required loci are correct; shade and label R.
8. **Matrix unknowns:** form an equation outside the matrix (12 + 4p = 4) for the method mark.
9. **Rearranging:** the answer must start with the subject ("must see d = …"); do not ISW a wrong further step.
10. **Factorise "completely":** take out every common factor; (3x − 1)(2x + 5)-type answers must be fully factorised.
11. **Quadratics:** if your three-term quadratic is wrong, method must be shown (substitution into the formula) to gain M1.
12. **Exact answers:** if decimals only are given where exact form is asked, no marks.
13. **Probability:** any probability greater than 1 in a calculation scores M0.
14. **Gradients from graphs:** a tangent must be drawn; a chord gradient scores 0 (Jun 25).
15. **Bounds:** show which bounds you use (e.g. 2135, 8.75, 0.75 for a lower bound of height, Jun 25).
16. **Percentage change:** a 4% increase reversed is ÷ 1.04, not × 0.96 ("3.5 seen from a 4% decrease – do not award any marks").
17. **Ratios:** give in simplest integer form (21 : 35 : 12).
18. **Factor theorem:** show f(a) = 0 explicitly with "= 0" and a conclusion.
19. **Answer on the answer line;** if your answer line differs from working, the answer line is marked.
20. **Accuracy:** 3 s.f. unless told; angles to 1 d.p. when asked.

3 NUMBER

SURDS WITHOUT A CALCULATOR

$(\sqrt{12} + \sqrt{300})/\sqrt{27} = (2\sqrt{3} + 10\sqrt{3})/(3\sqrt{3}) = 4$ (Jan 19). $3\sqrt{180} - 2\sqrt{245} = 18\sqrt{5} - 14\sqrt{5} = 4\sqrt{5} = \sqrt{80}$ (Jun 23). $\sqrt{700} - \sqrt{343} = 10\sqrt{7} - 7\sqrt{7} = 3\sqrt{7} = \sqrt{63}$ (Jun 26).

Rationalise: $(\sqrt{13} + \sqrt{11})/(\sqrt{13} - \sqrt{11}) = (\sqrt{13} + \sqrt{11})^2/2 = (24 + 2\sqrt{143})/2 = 12 + \sqrt{143}$ (Jan 21).

- LCM of 18, 30, 48 = 720 (Jun 19); HCF of 216 and 540 = 108 (Nov 21). Show prime factor trees or ladders.
- Remainder problems (Jun 21): N leaves 3 on division by 28 and by 120 ⇒ N = LCM(28, 120) + 3 = 843.
- Difference of two squares with numbers (Jan 21): $2^{36} - 1 = (2^{18} - 1)(2^{18} + 1) = 262\,143 \times 262\,145$.
- Mixed numbers: convert to improper fractions, show the common denominator, give the simplest mixed number.
- Standard form: $a \times 10^n$, $1 \le a < 10$; median of standard-form numbers: order them first (Jun 22).
- Golden ratio (Jun 22): $p = (1 + \sqrt{5})/2 \Rightarrow 1/p = 2/(1 + \sqrt{5}) = (\sqrt{5} - 1)/2 = p - 1$, shown by rationalising.

4 PERCENTAGES, RATIO, PROPORTION, BOUNDS

- Reverse percentage (Nov 20): after a 4% rise the price is \$3.64 ⇒ original 3.64 ÷ 1.04 = 3.50 ⇒ increase \$0.14.
- Profit targets (Jun 21): \$27 is 108% of cost ⇒ cost \$25 ⇒ 35% profit price \$33.75.
- Repeated change and loans (Jan 22): apply each year in order (× 1.04, then − 4000).
- Three-part ratio (Nov 20): a : b = 3 : 5 and a : c = 7 : 4 ⇒ 21 : 35 : 12.
- Chained proportion (Jun 25): $f = kg^2$, $g = m/h \Rightarrow f = 144/h^2$.
- Inverse proportion with roots (Jun 24): $y = k/\sqrt[3]{x}$; find k from the given pair, then substitute.

BOUNDS

Speed upper bound = $\text{distance}_{max} \div \text{time}_{min}$: 215 ÷ 3.375 = 63.7 km/h (Nov 21). Time upper bound = $\text{distance}_{max} \div \text{speed}_{min}$: 192.5 ÷ 61.5 = 3.13 h = 3 h 8 min (Nov 20).

Lower bound of an angle from an area formula (Jan 22): use A_{min} and a_{max}, b_{max} in $\sin C = 2A/(ab)$.

Differences of quantities (Jun 24): upper bound of $(P - Q) = P_{max} - Q_{min}$.

5 ALGEBRA

- Expand and factorise: $8xy^2 - 18xy = 2xy(4y - 9)$; $14x^2y^3 - 21x^5y^2 = 7x^2y^2(2y - 3x^3)$.
- Index laws: $(4a^2b^3)^3 = 64a^6b^9$; $(125p^9w^3)^{2/3} = 25p^6w^2$ (Nov 20).
- Algebraic fractions: factorise every numerator and denominator, then cancel or invert and multiply: $(3x^2 - 8x - 3)/(4x^2 - 36) = (3x + 1)(x - 3)/(4(x - 3)(x + 3)) = (3x + 1)/(4(x + 3))$ (Jun 19).
- Subject of a formula: clear fractions, collect the subject terms, factorise, divide; write "x = …".
- Completing the square (Nov 21): $15 - 28x - 7x^2 = 43 - 7(x + 2)^2$, so a = 43, b = −7, c = 2.
- Range of a quadratic function (Nov 20): $g(x) = 9 - 10x - 5x^2 = 14 - 5(x + 1)^2 \Rightarrow g(x) \le 14$.
- Factor theorem (Nov 24): f(5) = 250 − 225 − 55 + 30 = 0, so (x − 5) is a factor; $2x^3 - 9x^2 - 11x + 30 = (x - 5)(2x - 3)(x + 2)$.
- Explaining with algebra (Jan 21): $(2n^2 + 3n + 1)/(2n + 2) = (2n + 1)(n + 1)/(2(n + 1)) = (2n + 1)/2$, and 2n + 1 is odd, so no term is an integer.

6 EQUATIONS, INEQUALITIES, FUNCTIONS

- Linear equations with fractions: multiply every term by the LCM; "do not award if one term is not multiplied".
- Simultaneous equations: eliminate, substitute, give exact values (fractions) when they are not integers.
- Quadratic inequality (Nov 21): $6x^2 + 5x - 25 > 0 \Rightarrow (3x - 5)(2x + 5) > 0 \Rightarrow x < -2.5$ or $x > 5/3$.
- Double inequality (Jun 21): $-2x + 4 \le 3x - 6 < 2x + 2 \Rightarrow 2 \le x < 8$; show on a number line with closed/open circles.

- Linear programming regions: find the equation of each line, then write the three inequalities (Jun 22, Nov 23).
- Functions: fg(x) means g first; the range of $f(x) = x^2 - 2$ is $f(x) \geq -2$; excluded values make a denominator zero.
- Sequences: nth term substitution ($3n^2 + 11$ at n = 6, 9); show why two sequences share no terms (parity, Jun 26).

7 MATRICES

TOOLKIT

Add/subtract element by element; scalar multiply every element. Product AB: row of A × column of B; AB exists only if columns of A = rows of B.

Nov 20: A = (4 3 / −2 1), B = (−7 −2 / 1 3): A − B = (11 5 / −3 −2), AB = (−25 1 / 15 7).

Determinant of (a b / c d) = ad − bc. Inverse = (1/det)(d −b / −c a); no inverse if det = 0.

Unknown elements (Nov 21, Nov 23): multiply or add, then equate one element at a time outside the matrix.

8 GRAPHS, CALCULUS, KINEMATICS

- Gradient = $\Delta y/\Delta x$; equation y = mx + c; perpendicular gradient −1/m; distance between points by Pythagoras.
- Differentiate: $y = 2x^4 - 2/x \Rightarrow dy/dx = 8x^3 + 2/x^2$. Stationary points: dy/dx = 0.
- Speed–time graphs (Jan 23): gradient = acceleration (negative = deceleration); area = distance; set the area equal to the given distance to find u.
- Gradient of a curve at a point: draw a tangent, then rise ÷ run (Jun 25).
- Kinematics with calculus: v = ds/dt, a = dv/dt; at rest v = 0; reject negative times.
- Intersection of curve and line (Nov 24): equate, solve the quadratic; area of a triangle from the coordinates.

9 PROOFS: CONGRUENCE, SIMILARITY, ANGLES

CONGRUENCE (3 MARKS)

State three pairs of equal sides/angles, each with a reason, and the condition. **Jun 19:** BQ = QC (Q is the midpoint of BC); PQ = RC (opposite sides of a parallelogram are equal, AR = RC); QR = BP (opposite sides of a parallelogram, AP = PB) ⇒ congruent (SSS).

Allowed conditions: SSS, SAS, ASA/AAS, RHS. Two correct pairs with reasons earns M2; the conclusion needs all three.

SIMILARITY (3 MARKS)

Two pairs of equal corresponding angles with reasons: e.g. ∠BAE = ∠CDE (angles in the same segment), ∠AEB = ∠DEC (vertically opposite angles) ⇒ triangles ABE and DCE are similar (Jun 21).

Similar by sides (Nov 20): show all three ratios of corresponding sides are equal.

CIRCLE THEOREMS AND CHORD RULES

Angle at centre = 2 × angle at circumference · angle in a semicircle = 90° · angles in the same segment are equal · opposite angles of a cyclic quadrilateral sum to 180° · tangent ⊥ radius · tangents from a point are equal · alternate segment theorem.

Intersecting secants (Nov 21): EA × EB = ED × EC ⇒ 7.5(7.5 + x) = 6 × 14 ⇒ x = 3.7. Intersecting chords: AE × EC = BE × ED.

- Polygons: exterior angle = 360/n. Exterior 9° ⇒ 40 sides (Jun 19); interior 176° ⇒ 90 sides (Jan 21); angle sum 8280° ⇒ n = 48, exterior 7.5° (Jan 22).
- Proving the angle sum of a triangle (Jun 22): use alternate angles with a parallel line, giving each reason.

10 CONSTRUCTIONS AND LOCI

Locus	Construction (keep all arcs)
fixed distance from a point	circle or arc, compass set to the distance (scale it first)
equidistant from two points	perpendicular bisector: two pairs of equal-radius arcs from each point
equidistant from two lines	angle bisector: arc from the vertex cutting both arms, then equal arcs from those points
fixed distance from a line	parallel line at that distance

- "Closer to B than to C" is the side of the perpendicular bisector containing B; "closer to AB than to AC" is the side of the angle bisector next to AB.
- Region R (Jan 19, Nov 20, Jan 23, Jun 26): construct every locus, then shade only the part satisfying all conditions; label it R.
- Scale drawings (Jan 22, Jan 23): convert metres to centimetres with the scale before setting the compasses.
- Symmetry (Jun 19, Jun 21, Nov 21): one line of symmetry vs rotational symmetry of order 2 – check both after shading.

11 MENSURATION, SIMILARITY, CIRCLES

- No formulae sheet: learn volume of cone $\frac{1}{3}\pi r^2 h$, sphere $4/3\pi r^3$, curved surface of cone πrl, sphere surface $4\pi r^2$, cylinder $2\pi rh$, prism = cross-section × length.
- Sectors: arc = $\theta/360 \times 2\pi r$; area = $\theta/360 \times \pi r^2$. From the area find r, then the perimeter (Jun 22).
- Similar shapes: length k, area k^2, volume k^3. Volume up 33.1% ⇒ $k^3 = 1.331$ ⇒ k = 1.1 ⇒ area × 1.21 (Jun 22). Volume down 43% ⇒ $k^3 = 0.57$ (Nov 23).
- Similar triangles (Nov 21): EB/DC = AE/AD ⇒ EB = 12.6 × 8/11.2 = 9; AC = 8.5 × 11.2/8 = 11.9 cm.
- Composite solids: spheres in a cuboid, cylinder in a cube, hemisphere on a cone – subtract or add the parts.

12 TRIGONOMETRY AND VECTORS

- Right-angled triangles: SOH CAH TOA and Pythagoras; angles of elevation from a raised point need the height difference.
- Non-right triangles: sine rule, cosine rule (smallest angle is opposite the shortest side, Nov 21), area $\frac{1}{2}ab \sin C$ — all to be learnt.
- 3D (Jun 21): angle between AE and the base uses AX, half the diagonal of the base.
- Bearings (Jun 25): draw north lines, find the angle at L (125° − 50° = 75°), use the cosine then sine rule.
- Vectors: column vector magnitude $\sqrt{(x^2 + y^2)}$; a = kb means parallel; express a path through known vectors; ratio questions by equating coefficients (Jun 19, Jun 24, Nov 25).

13 STATISTICS

- Grouped mean: Σ(midpoint × f) ÷ Σf; give to the accuracy asked. Discrete mean: Σ(x × f) ÷ Σf.
- Missing frequency from the mean (Nov 25): Σxf = mean × Σf, form a linear equation in x.
- Combined means (Jun 24, Jun 25): total = mean × number; subtract the known total.
- Mean and median conditions (Jan 21, Nov 21, Jun 23): write expressions for both and form an equation.

- Histograms: frequency density = f ÷ width; complete missing bars and table entries; probabilities from areas.
- Bar charts and pie charts: angle = fraction × 360°; height scales with frequency.

14 PROBABILITY AND SETS

- Probabilities sum to 1: solve for x in a table first ($2x - 0.1 + 3x + 0.2 + \ldots = 1$).
- Trees: "only one" = P(win, not) + P(not, win); compare events by computing both (Nov 24).
- Without replacement: denominators fall; unknown numbers lead to a quadratic (Nov 21).
- Not the same colour = 1 − P(same colour) (Jun 19).
- Sets: $A \cup B$, $A \cap B$, A', , n(A); Venn diagrams from the centre outwards; $n(R \cap C')$ is "R only" (Jun 21).

15 FINAL A* CHECKLIST

1. No formulae sheet: learn the mensuration, trigonometry and quadratic formulae.
2. Surds and fractions: every stage written, no calculator values.
3. Proofs: each statement with its reason, then SAS/ASA/RHS/SSS or AA.
4. Constructions: arcs visible; region R shaded and labelled.
5. Matrices: check orders; equations outside the matrix for unknowns.
6. Algebra: factorise completely; rearranged answers begin with the subject.
7. Bounds: correct bound for each part of the formula.
8. Percentages: reverse changes divide by the multiplier.
9. Circle theorems: underlined words in every reason.
10. Keep moving: under one minute per mark; leave 5 minutes to check.